PROJECT REPORT

Organizational Employee Dashboard

Using Microsoft Power BI

Project Name	Organizational Employee Dashboard
Tool Used	Microsoft Power BI Desktop
Institute Name	Besant Technologies, Chennai
Course	Data Analytics (Excel, SQL, Power BI, Python)
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Submitted To	
Date of Submission	

Acknowledgement

I would like to express my sincere gratitude to **Besant Technologies** for giving me the opportunity to learn Data Analytics and to work on this practical project.

I thank my trainer for the clear guidance, patient support and valuable feedback throughout the Power BI sessions. The training on data modelling, DAX, visualisation and report design helped me to complete this project successfully.

I also thank my family, friends and classmates for their constant encouragement and support during the course.

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4. Introduction to Power BI

Power BI is a business analytics tool from Microsoft that turns raw data from many sources into interactive, easy-to-understand reports and dashboards. It helps people to see trends, compare values and make data-driven decisions without needing to write complex code.

Power BI Desktop - the free Windows application used in this project - lets an analyst connect to data, clean and shape it with Power Query, build relationships between tables, write calculations with DAX and design visual reports.

Main components

- **Power BI Desktop** - for building data models and report pages.
- **Power Query Editor** - for cleaning and transforming data (ETL).
- **Data Model** - tables, relationships and measures.
- **Visuals** - bar charts, pie charts, donut charts, cards, tables and slicers.
- **Power BI Service** - for publishing and sharing reports online.

Why Power BI?

- Easy drag-and-drop report design with interactive filtering.
- Connects to Excel, CSV, SQL databases and many other sources.
- Fast, professional dashboards that update when the data changes.

5. Introduction to the Project

The **Organizational Employee Dashboard** is an interactive Power BI report built on an Employee Sample Data workbook. It analyses the salary, bonus and location details of employees of an organization across departments, business units, countries, cities and ethnic groups.

Objectives

- Build a proper data model from the Employee Sample Data using fact and dimension tables.
- Summarise the total, maximum and minimum annual salary using KPI cards.
- Compare salary by department, unit, country and city.
- Give users slicers to filter by Department, Unit, Country and City.
- Create a Male Employee Dashboard and separate analysis pages for each ethnicity.
- Use buttons and bookmarks so users can move easily between pages.

Scope of the Dashboard

The report contains one main page (Male Employee Dashboard) and four drill-through analysis pages - **Latino**, **Caucasian**, **Black** and **Asian** Ethnicity Analysis. Each page has a back button that returns the user to the main dashboard.

6. Data Model

A star-schema style model was created in Power BI Desktop. One central **fact table** holds the organizational records, and smaller **dimension tables** describe the employees, departments, ethnicity and locations. Relationships were created between the fact table and each dimension table.

Table	Type	Purpose
Organization Data	Fact table	Central records - salary, bonus % and keys to other tables
Employee Details	Dimension	Employee-level details such as gender
Department and Units	Dimension	Department and business unit of each employee
Ethnicity	Dimension	Ethnic group of each employee
Country and City	Dimension	Location - country and city

Benefits of this model

- Removes repeated data and keeps the model clean.
- Slicers on a dimension table filter every visual connected to the fact table.
- Makes reports faster and easier to maintain.

7. Concepts Used

Concept	Where it is used
Data import & Power Query	Loading and cleaning the Employee Sample Data workbook
Data modelling & relationships	Fact and dimension tables linked together
Measures / aggregations	Sum, Maximum and Minimum of Annual Salary; Sum of Bonus %
Card visuals	Total, Maximum and Minimum Annual Salary KPIs
Clustered column chart	Sum of Annual Salary by Country; Total Annual Salary by Department
Bar chart	Total Annual Salary by City (Caucasian page)
Pie chart	Salary by Department; Salary by Country (Latino page)
Donut chart	Sum of Annual Salary by Unit (Black page)
Matrix / table with conditional formatting	Employee Salary & Bonus Details - green data bars/colour on Bonus %
Slicers	Department, Unit, Country and City filters
Buttons & bookmarks	Back arrow and navigation buttons between pages
Visual-level filters	Gender = Male and Ethnicity filters on each page

8. Dashboard Pages and Output Screenshots

8.1 Male Employee Dashboard (Main Page)

This is the main page. Slicers at the top (Department, Unit, Country, City) filter the page. Three KPI cards show Total Annual Salary (55M), Maximum Annual Salary (258K) and Minimum Annual Salary (40K). Below are a column chart (salary by country), a pie chart (salary by department) and a table with total salary and bonus % per department.

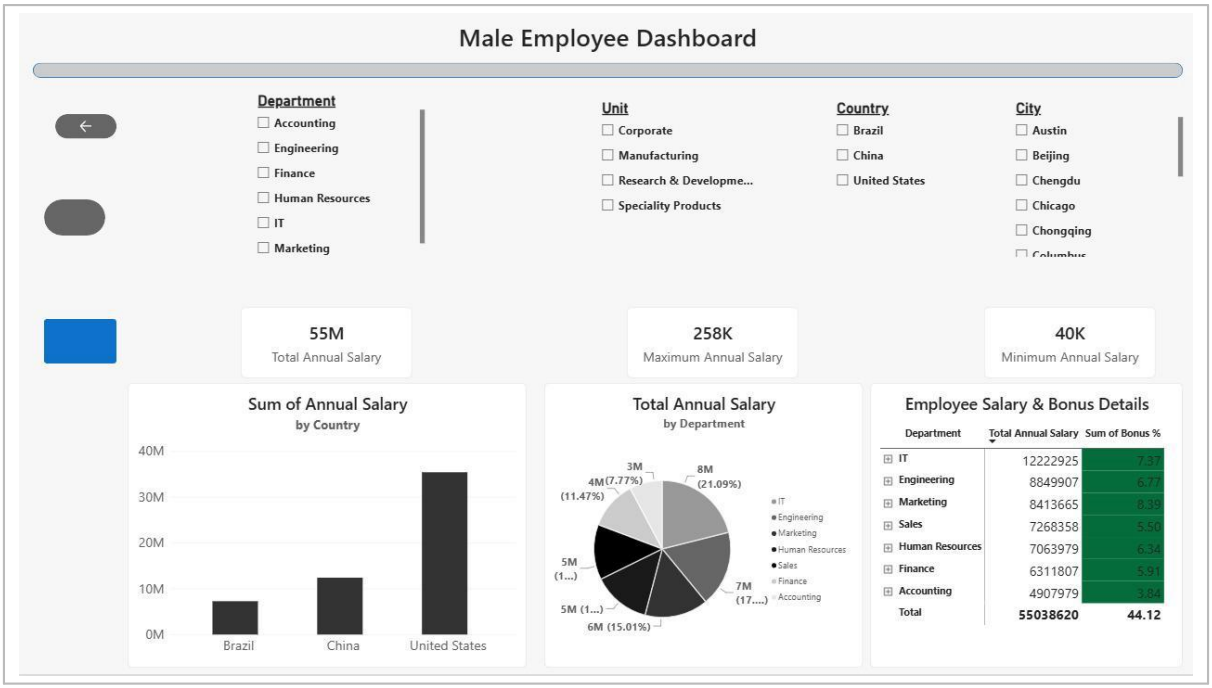


Figure 1: Male Employee Dashboard

Department-wise salary and bonus (from the table visual)

Department	Total Annual Salary	Bonus %
IT	12,222,925	7.37
Engineering	8,849,907	6.77
Marketing	8,413,665	8.39
Sales	7,268,358	5.50
Human Resources	7,063,979	6.34
Finance	6,311,807	5.91
Accounting	4,907,979	3.84
Total	55,038,620	44.12

8.2 Latino Ethnicity Analysis

A pie chart of Total Annual Salary by Country shows that Latino employees are located in Brazil and the United States. Brazil holds the larger share of the salary total.

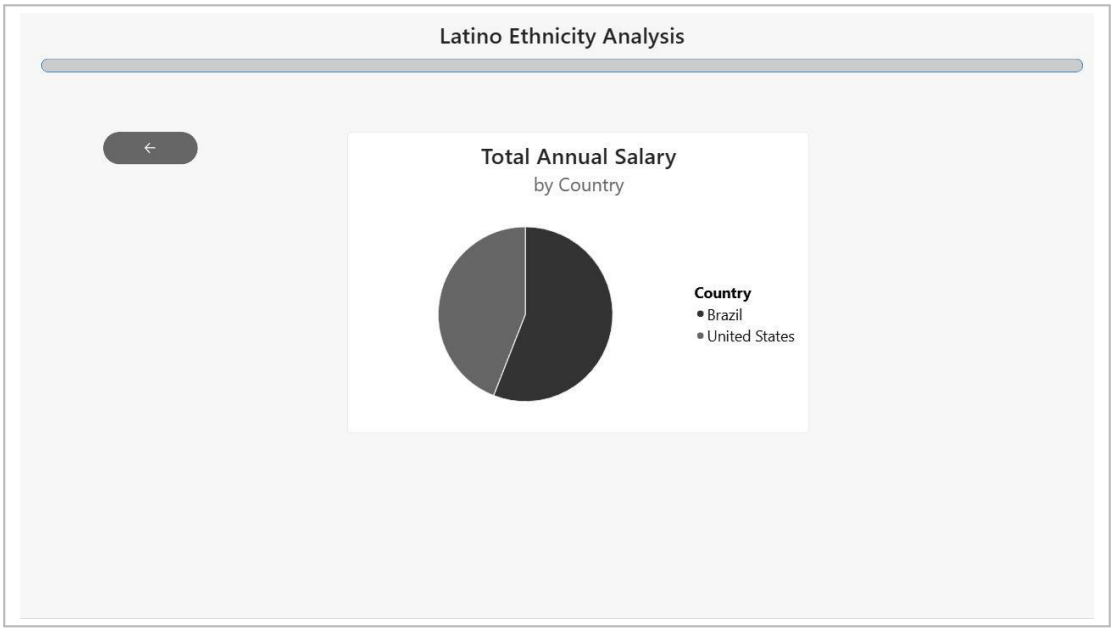


Figure 2: Latino Ethnicity Analysis - Total Annual Salary by Country

8.3 Caucasian Ethnicity Analysis

A bar chart of Total Annual Salary by City. Miami (about 3.5M) and Seattle (about 3.3M) lead, followed by Columbus, Phoenix and Chicago, while Austin is lowest at about 1.0M.

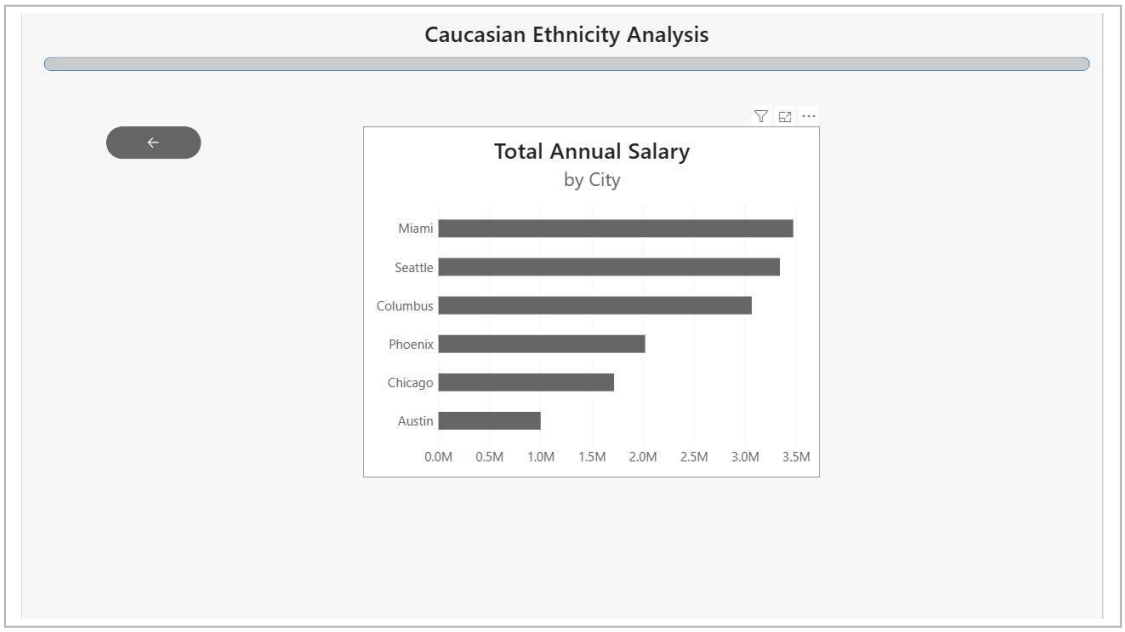


Figure 3: Caucasian Ethnicity Analysis - Total Annual Salary by City

8.4 Black Ethnicity Analysis

A donut chart of Sum of Annual Salary by Unit. The salary is split equally across Corporate, Manufacturing, Research & Development and Speciality Products - about 4M (25%) each.

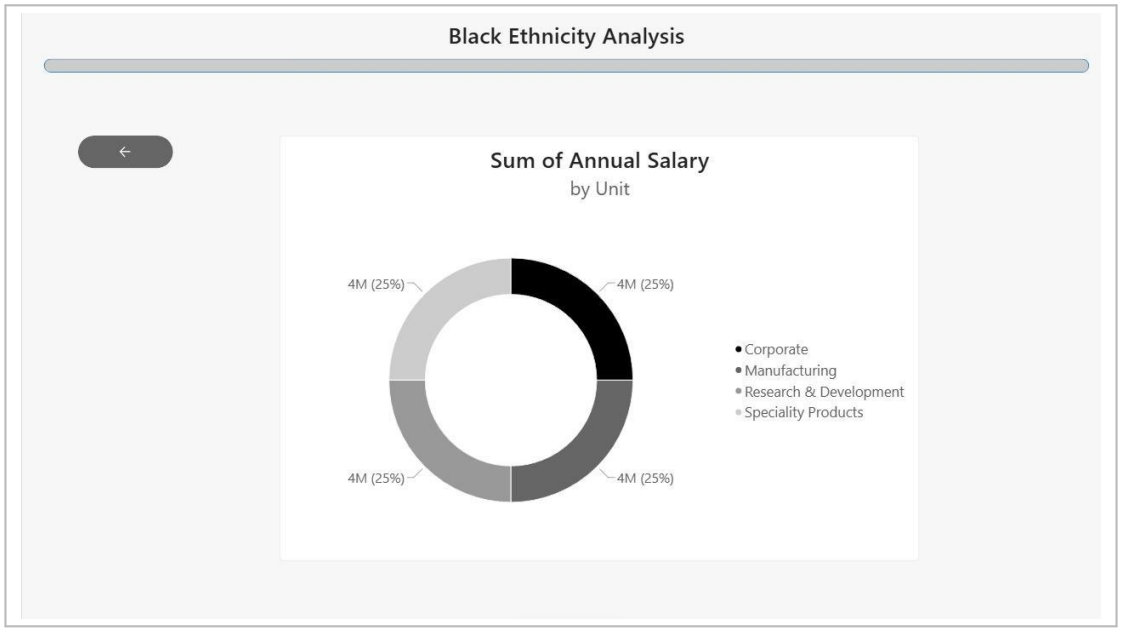


Figure 4: Black Ethnicity Analysis - Sum of Annual Salary by Unit

8.5 Asian Ethnicity Analysis

A column chart of Total Annual Salary by Department with data labels. IT (4.8M) and Marketing (4.7M) are highest; Accounting (2.2M) is lowest.

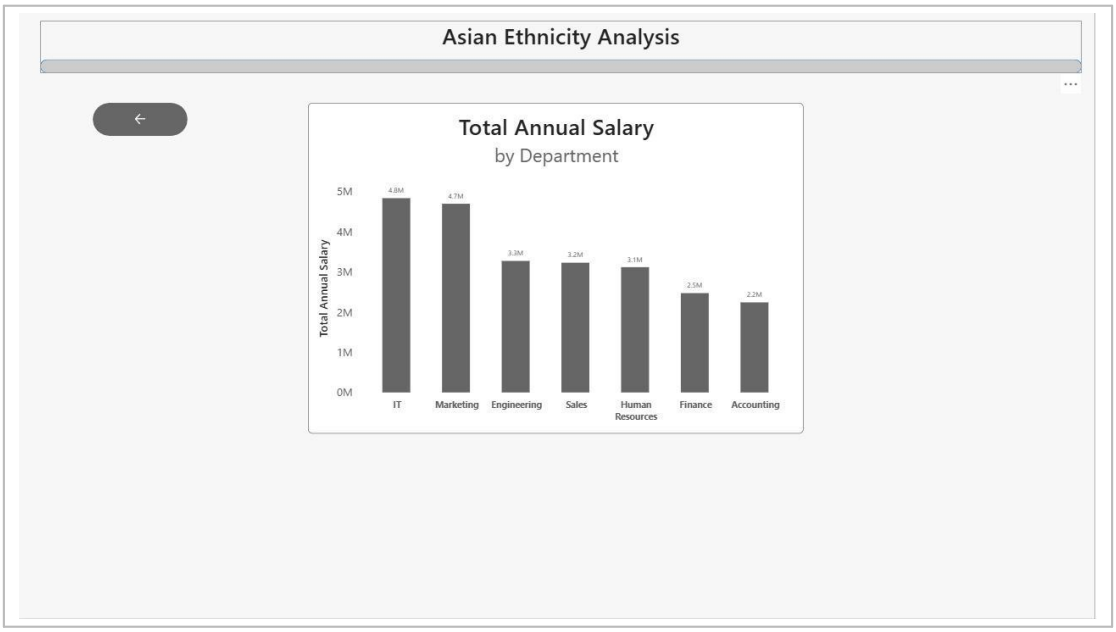


Figure 5: Asian Ethnicity Analysis - Total Annual Salary by Department

9. Key Insights

- The total annual salary of male employees is about **55M**, with a maximum of 258K and a minimum of 40K.
- The **United States** accounts for the largest share of salary (about 35M), far above China and Brazil.
- **IT** is the highest-paid department (about 12.2M) and also has a high bonus of 7.37%; **Accounting** is the lowest (about 4.9M, bonus 3.84%).
- **Marketing** has the highest bonus percentage (8.39%) among departments.
- Among Asian employees, **IT** and **Marketing** lead in salary; among Caucasian employees, **Miami** and **Seattle** lead by city.
- For Black employees, salary is evenly distributed across all four business units.

Note: the values above are read from the dashboard visuals and are rounded where the charts show rounded labels.

10. Conclusion

The Organizational Employee Dashboard shows how Power BI can turn a plain employee data sheet into an interactive, easy-to-read analysis. By building a clear data model, using the right visuals and adding slicers, buttons and bookmarks, the dashboard lets any user explore salary and bonus patterns by department, unit, country, city and ethnicity in a few clicks.

Through this project I gained hands-on experience in data modelling, relationships, visual selection, formatting, filtering and navigation design in Power BI. In the future, the dashboard can be extended with female employee pages, year-wise trends, DAX measures for averages and growth, and can be published to the Power BI Service for sharing.

11. Bibliography

- Microsoft Learn - Power BI documentation: learn.microsoft.com/power-bi
- Besant Technologies - Data Analytics course material and class notes
- Employee Sample Data workbook provided for the project